

25 July 2026

Editorial Office

Neurocomputing

Re: Research article submission, “Distribution-free certification of learned progressive policies for retrieval-augmented generation under knowledge drift”

Dear Editor,

On behalf of all authors, we submit this manuscript for consideration as a Research Article in *Neurocomputing*.

Dynamic retrieval-augmented generation (RAG) learns when to retrieve, rerank, stop, or escalate to a stronger model. Such mechanisms improve average accuracy and efficiency, but the learned score itself does not bound the population risk of the policy selected for deployment. This problem becomes more serious when many policy thresholds are searched or the external knowledge distribution changes.

Our manuscript introduces ContractRAG, a learning-and-certification method for progressive RAG inference. Per-rung sufficiency models use retrieval and generation signals to control adaptive evidence acquisition. Composable risk vouchers order policies generated from a rewrite lattice. A separate distribution-free fixed-sequence algorithm then certifies the selected policy independently of candidate count and score quality. We prove finite-sample contract validity and cost consistency. We also develop a restart-mixture e-process with change-time-independent detection delay, constant-time updates, and a lifetime guarantee across repeated recertification.

The evaluation uses HybridQA, CRAG, ASQA, and QAMPARI, four model families, and two serving backends. It includes ablations of score features and policy families, calibration-size scaling, dense candidate grids, strict and joint contracts, audit rates, and controlled drift. Across 1,000 calibration draws, selected policies exceed their contracts in at most 0.8% of draws under a 10% error budget, whereas empirical and Bayesian selection reach 64%. Model cost falls by up to $20\times$, or $10\text{--}18\times$ after auditing.

The paper connects learned reranking, iterative retrieval, knowledge refinement, and uncertainty-aware retrieval through a general selection and certification layer that incorporates them as policies or rungs. We believe this combination of learned adaptive computation, statistical learning guarantees, drift adaptation, public-benchmark experiments, and mechanism-focused ablations fits the journal’s interest in learning methods, computational learning theory, and resource-aware neural systems.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and its submission. The authors declare no competing financial interests or personal relationships that could have influenced the work.

Thank you for considering our submission.

Sincerely,

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